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ORIGINAL REPORT

Factors for Hematopoietic Toxicity of Carboplatin: Refining the Targeting of Carboplatin Systemic Exposure

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ABSTRACT

Purpose

Area under the curve (AUC) dosing is routinely carried out for carboplatin, but the chosen target AUC values remain largely empirical. This multicenter pharmacokinetic-pharmacodynamic (PK-PD) study was performed to determine the covariates involved in the interindividual variability of carboplatin hematotoxicity that should be considered when choosing individual target AUCs.

Patients and Methods

Three hundred eighty-three patients received carboplatin as part of established regimens. A semi-physiologic population PK-PD model was applied to describe separately the time course of absolute neutrophil and platelet counts using NONMEM software. The plasma ultrafiltrable carboplatin concentration (C_{Carbo}) was assumed to inhibit the proliferation of blood cell precursors through a linear model: drug effect = slope × C_{Carbo}. The slope corresponds to the patients' sensitivity to carboplatin hematotoxicity. The relationships between the patients' sensitivity to the neutropenic or thrombopenic effects of carboplatin and various covariates, including associated chemotherapies, demographic, biologic, and pharmacogenetic data, were studied.

Results

The sensitivity of carboplatin-induced thrombocytopenia decreased in the case of concomitant paclitaxel chemotherapy (slope decreased by 24%), whereas it increased with coadministration of etoposide and gemcitabine (slope increased by 45% and 133%, respectively). For neutropenia, the sensitivity increased when carboplatin was combined with other cytotoxics (slope increased by 76%).

Conclusion

This study provides useful information to clinicians to better estimate the hematopoietic toxicity of carboplatin and thus choose more rationally carboplatin target AUCs as a function of pretreatment or concomitantly administered chemotherapies. For example, an AUC of 5 mg/mL · min is associated with a risk of grade 3 or 4 thrombocytopenia of 2% in combination with paclitaxel versus 38% with gemcitabine in a non-pretreated patient.

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INTRODUCTION

Several formulas are used for individual carboplatin dosing.[1-3] All are based on the choice of a target area under the curve (AUC) of plasma ultrafiltrable carboplatin concentrations versus time. The optimal carboplatin AUC may be defined as the plasma exposure that maximizes the likelihood of a response while maintaining toxicity at a manageable level. The dose-limiting toxicity of carboplatin is myelosuppression, in particular thrombocytopenia. Although some studies have proposed optimal AUCs of carboplatin in combination with cyclophosphamide, etoposide,[4] or paclitaxel,[5] they are all based on a limited number of patients. Moreover, the target AUC values were derived from pharmacokinetic (PK) -pharmacodynamic (PD) relationships between the likelihood of specific grades of toxicity to platelet counts or absolute neutrophil counts (ANC) and AUC values. These methods of analysis did not consider either the entire time course of platelet counts or ANC or the patients' characteristics that may be involved as covariates. As a result, the choice of the target AUC for a particular patient currently remains largely empirical.

The objective of this prospective multicenter study was to identify relevant covariates linked to the interpatient variability of carboplatin hematotoxicity to better evaluate the risks and thus improve individual carboplatin dosing. The study was conducted with 383 patients receiving carboplatin as part of established monotherapy or combination regimens. The relationship between carboplatin PK and hematotoxicity was described by using a population PK-PD approach based on a semi-mechanistic model proposed by Friberg et al[6] for several cytotoxics. This modeling approach considers simultaneously the entire time course of platelet or ANC, the carboplatin plasma concentrations versus time, and the patient characteristics. The identification of covariates that modulate the hematotoxicity was performed using a model-building data set composed of 276 patients (training data set). The effects of these covariates were then evaluated prospectively using the data from 107 other patients (validation data set).

PATIENTS AND METHODS

Patients

Thirteen study centers participated in the trial (Appendix, online only). The protocol was approved by the ethical committee of Toulouse, and informed written consent was obtained from each patient. Four hundred patients were recruited between 2005 and 2008. The inclusion criterion for participation in the study was to be treated with carboplatin as part of an established regimen (Table 1).

Carboplatin Administration, Blood Sampling, and Population PK Analysis

Carboplatin was administered as a daily 30- or 60-minute intravenous infusion in 5% dextrose at doses calculated according to either Calvert formula combined with the Cockcroft-Gault equation, Chatelut formula, or body-surface area, depending on the centers and the regimen.

CBCs were done weekly after the first and second cycles. Albuminemia, proteinemia, bilirubinemia, and uremia were measured at inclusion. Four blood samples were obtained at time 0 (before administration), 5 minutes before the end of the infusion, and at 1 and 4 hours after the end of the infusion. Predictions of individual carboplatin PK parameters were obtained as described in detail[7] by Empirical Bayes Estimates using the POSTHOC option of the NONMEM program[8] (version VI, level 1.0, Icon Development Solutions, Ellicot City, MD). This analysis enables a complete plasma carboplatin concentration versus time profile to be generated for each patient. In three centers, no PK exploration was planned. For these patients with no PK blood sampling, carboplatin clearance was calculated according to the Thomas modified formula based on creatininemia, plasma cystatin C, age, body weight, and sex.[7]

Basic Population PK-PD Model for Absolute Neutrophil and Platelet Counts

Patients were split into two groups for the analysis: one group to build the model using 280 patients (training data set) and one group for model validation using the rest of the patients (validation data set). The structural model used in the present analysis to describe the whole platelets and neutrophils time course was that proposed by Friberg et al[6] (Fig 1A). The model is described in the Appendix (online only).

For both ANC and platelet counts modeling, the rate constants: k_{prol} (proliferation of progenitor cells), k_{tr} (transit between compartments), and k_{circ} (physiological elimination of circulating cells) were assumed to be equal to a single rate constant k as in the original model. Three parameters were used to characterize the system: Base, the mean transit time through the maturation delay chain ($MTT = 4/k$), and γ as the power factor for the feedback mechanism. The carboplatin concentration in the central compartment (C_{Carbo}, corresponding to the plasma ultrafiltrable concentration) was assumed to induce cell loss from the progenitor cells compartment (Prol). The drug effect, E_{drug}, was proportional to C_{Carbo}: $E_{drug} = \text{slope} \times C_{Carbo}$. The interindividual variability was estimated for base, MTT, slope, and γ (for platelets only) and inter-occasion (ie, inter-cycles) variability was added to slope. A log-normal distribution of individual PK-PD parameters was assumed.

Table 1. Characteristics of the 383 Patients (female/male: 279/104)

Characteristic	Mean	Minimum	Maximum
Total no. of patients	383		
Sex			
Male	104		
Female	279		
Dose of carboplatin administered, mg	543	170	1200
BSA, m^2*	1.7	1.3	2.37
Weight, kg	65	40	137
Height, cm	164	146	187
Age, years	60	21	87
Proteinemia, g/L	69	40	88
Albuminemia, g/L	36	13	50
Bilirubinemia, μmol/L	7.3	1	26
Uremia, mmol/L	5.7	1.2	42
Serum creatinine, μmol/L	76	25	433
Plasma cystatin C, mg/L	0.89	0.45	3.49
Platelet baseline, $\times 10^9$/L	362	105	1031
Neutrophil baseline, $\times 10^9$/L	5.645	1.357	25.370
GSTP1 polymorphisms			
AA	167		
AG	174		
GG	42		
CC	322		
CT	58		
TT	3		
ERCC1 polymorphisms			
CC	67		
CT	187		
TT	129		
GG	201		
GT	159		
TT	23		
Primary tumor site			
Ovary	164		
Uterus	47		
Lung	46		
Soft tissues	29		
Others	79		
Unknown	18		
Previous chemotherapy			
Yes	137		
No	246		
Multiple-dose corticosteroid administration			
Yes	82		
No	301		
Associated concomitant chemotherapy			
None	72		
Paclitaxel 175 mg/m^2†	205		
Etoposide 120 mg/m^2 on days 1, 2, 3†	32		
Gemcitabine 1,000 mg/m^2 on days 1, 8†	20		
Others (fluorouracil, docetaxel, vinorelbine, trastuzumab, doxorubicin, topotecan, fotemustine, vinblastine)	54		
Performance status			
0	161		
1	168		
2	40		
3	14		

Abbreviations: BSA, body-surface area; *GSTP1*, glutathione-*S*-transferase P1; *ERCC1*, excision repair cross-complementing 1.
*Calculated according to the Dubois method.
†Median dose.

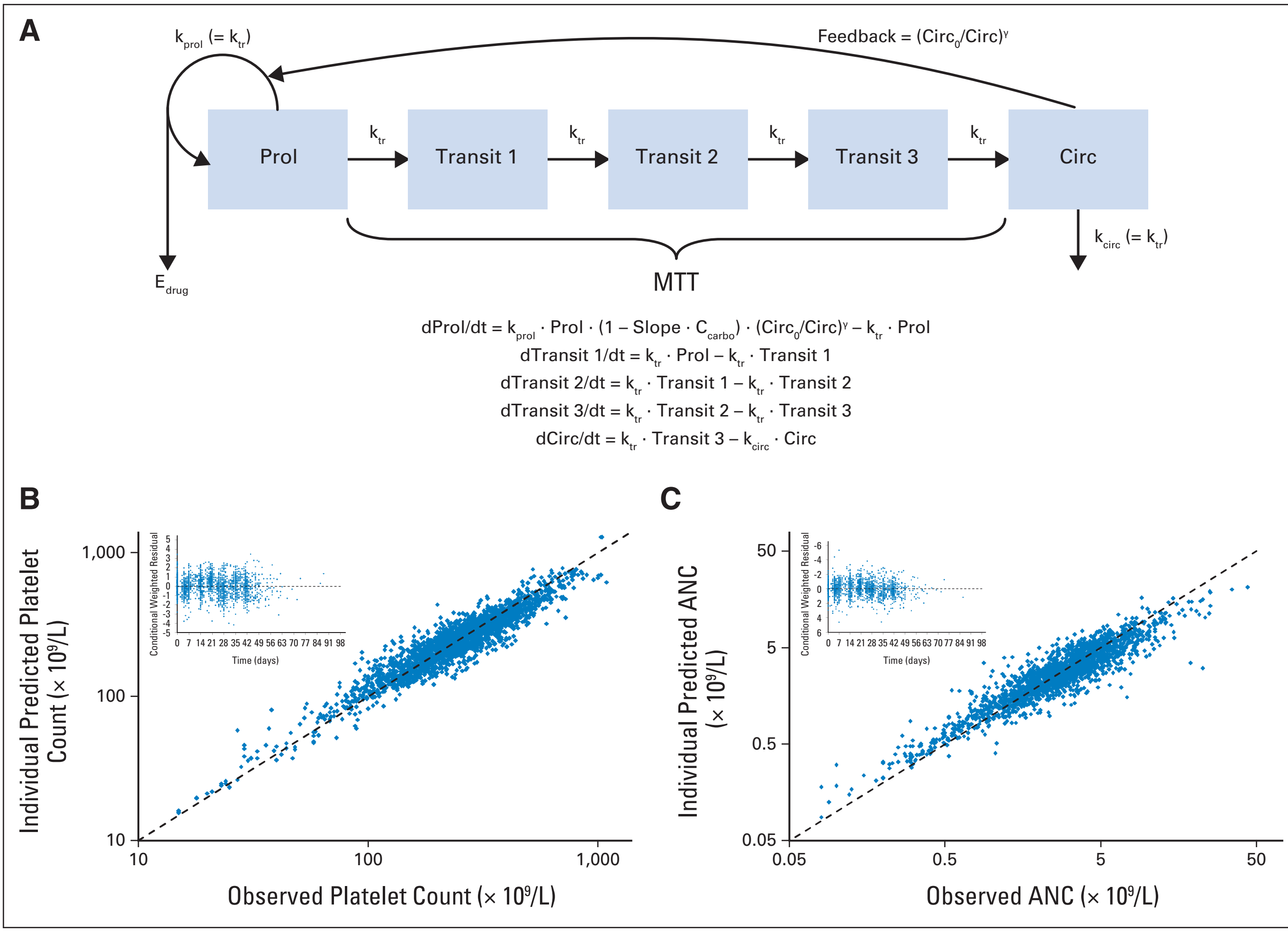

Fig 1. Goodness-of-fit plots for the basic model (without any covariates) obtained with the training data set (n = 276 and = 249 for platelet count and absolute neutrophil count [ANC] respectively). (A) Pharmacokinetic-pharmacodynamic model describing myelosuppression after carboplatin administration (modified from Friberg et al[6]). (B) Individual predicted versus observed platelet count; insert: conditional weighted residuals of the platelet model versus time. (C) Individual predicted versus observed ANC; insert: conditional weighted residuals of the neutrophil model versus time. MTT, mean transit time. Prol, progenitor cells; Circ, circulating cells; $Circ_0$, basal cell count.

The observed neutrophil and platelet data were log-transformed before the analysis, and an additive error model was used to obtain the residual variability. The first order conditional estimates method in NONMEM was applied for the estimation of the model parameters.

Covariate Analysis

The relationships between the three PK-PD parameters (ie, base, MTT, and slope) and 18 patient covariates were explored by using the training data set. The covariates studied were age, body-surface area (BSA), body weight (BW), height, sex, performance status, albuminemia (ALB), proteinemia, uremia, bilirubinemia, lymphocyte count at day 5, multiple-dose (> 4 days) corticosteroid administration (CORT), monochemotherapy versus combination with other cytotoxics (paclitaxel, etoposide, gemcitabine, or others), previous chemotherapy or not, and four single-nucleotide polymorphisms (SNP) corresponding to either glutathione-*S*-transferase P1 (*A313G*, and *C340T*) or excision repair cross-complementing 1 (*C353T*, and *G8092T*). First, predictions of the individual PK-PD parameters obtained from the basic model were plotted against each covariate. The covariates for which this preliminary screening revealed a significant relationship were further tested in NONMEM. The covariate selection and the obtainment of the final model from the intermediate model are described in detail in the online-only Appendix.

Evaluation of the Final Population PK-PD Models

Standard goodness-of-fit plots using conditional weighted residuals[9] were done at each step of the model building. Moreover, visual predictive checks (VPCs) were carried out using the validation data set: 1,000 replicates of the study design were simulated using the final models. Simulated platelet counts and ANCs (summarized by their median and 90% prediction interval at each time) were then compared with the actual observed values in the patients comprising the validation data set. In cases of agreement between the model and the observations, both training and validation data sets were pooled, and the model was refined by re-estimating all the model parameters. VPCs were again done for this final model using the global data set (n = 383 for platelet counts, and n = 338 patients for ANCs), as well as for each patient subgroup defined by significant covariates.

For additional evaluation of the effects of covariates, the final PK-PD covariate models obtained from the training data set were applied to the validation data set and all the covariate effects were re-estimated.

Prediction of Carboplatin Hematotoxicity

Individual platelet counts and ANC time courses were simulated in NONMEM using the final PK-PD covariate model estimated from the whole data set and 5,000 virtual patients for different levels of plasma carboplatin AUC (ranging from 4 to 8 mg/mL · min by steps of 0.5 mg/mL).

RESULTS

Patient Population and Treatment

From the 400 included patients, 29 did not have a PK exploration and were allocated to the validation data set. Thirteen patients did not receive carboplatin, and four patients were treated with dose-intensive carboplatin regimens (ie, a target AUC of 24 mg/mL · min). Exclusion of these 17 patients led to a training and a validation data set of 276 and 107 patients, respectively. For the ANC analysis, data from 45 patients (27 and 18 in the training and validation data sets, respectively) were excluded because they received granulocyte colony-stimulating factors. The main patient characteristics and carboplatin regimens are shown in Table 1.

Final Population PK-PD Model for Platelet Counts and ANCs

The population PK-PD models fully described the time course of both platelets and neutrophils after carboplatin administration, as shown by standard goodness-of-fit plots (Fig 1). Observations versus individual prediction graphs showed that there was no bias due to the structural model except a slight overprediction of the lowest ANC. The preliminary screening of covariates and further testing in NONMEM using the forward/backward procedure led to the final population PK-PD models described next for platelet counts and ANCs based on the training data set.

Platelets. Basal level (base) was significantly lower in patients pretreated with chemotherapy: 21% (± 7%) decrease on average (± 95% CI). Importantly the thrombocytopenia sensitivity (slope) was dependent on the carboplatin regimen; considering monochemotherapy as the reference, the slope was significantly increased by 156% (± 64%) in combination with gemcitabine, and by 44% (± 42%) in combination with etoposide. In contrast, the sensitivity significantly decreased by 21% (± 13%) in combination with paclitaxel. The number of patients treated with other concomitant chemotherapies was small (54 patients), and they were taken all together. There was no significant difference between the thrombocytopenia sensitivity of the patients with other concomitant chemotherapies and those who received carboplatin as monochemotherapy.

Neutrophils. Basal level (base) was associated with ALB and CORT (typical value of base $= 4.47 \times (\mathrm{ALB}/37)^{-0.72(\pm\ 0.41)} \times 1.41(\pm 0.19)^{\mathrm{CORT}}$). The MTT and neutropenia sensitivity (slope) were dependent on the carboplatin regimen; considering monochemotherapy as the reference, the MTT was significantly decreased by 49% (± 32%) and slope was increased by 86% [± 13%] when carboplatin was combined with another cytotoxic drug, whatever the combined drug. Indeed, as shown in Figure 2, the individual slope values for the ANC (ie, empirical Bayes estimates from the basic model applied to the whole data set) were not significantly different between carboplatin combination regimens.

These covariate models were applied to the validation data set: All covariates remained significant except previous chemotherapy on basal level of platelets and combination with another cytotoxic drug on the MTT for the ANC.

Finally, final PK-PD models corresponding to either platelet counts or ANCs were applied to the whole data set (ie, combined training and validation data set): The significance of previously identified covariates and PD parameters were confirmed. The comparison between the covariate coefficients obtained from the whole data set (Table 2) and the above values obtained from the training data set illustrates the consistency between the analyses.

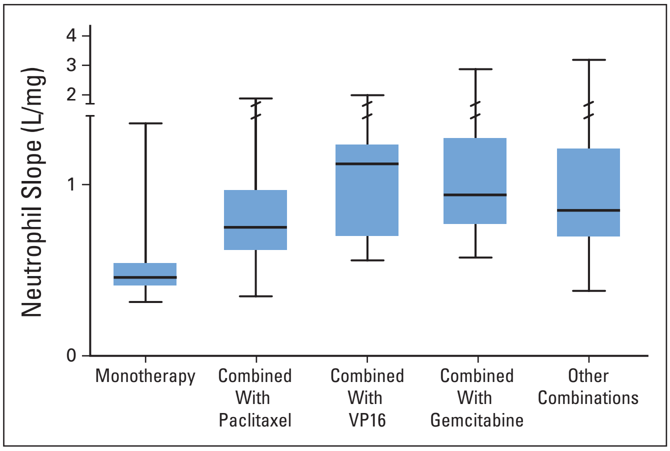

Fig 2. Box plots of individual slope values for absolute neutrophil counts obtained from the final model on the whole data set according to different regimens. VP16, etoposide.

Evaluation of the Final Population PK-PD Models

VPCs were done using the validation data set for both population models developed for neutrophils and platelets from the training data set. They were also performed for the final models estimated from the whole data set. All VPCs were satisfactory (Fig 3 for the whole data set and Appendix Fig A1, online only, for the training data set). These VPCs were also done for each subgroup according to the main significant covariates (data not shown). Overall, the results of model evaluation were good, which highlights the ability of the final models to describe the data.

Prediction of Carboplatin Hematotoxicity From the Final PK-PD Models

The percentage of patients experiencing grade 3 or worse thrombocytopenia or grade 4 neutropenia after cycle 1 is shown in Figure 4 for each patient subgroup defined according to carboplatin regimen.

DISCUSSION

Several retrospective studies indicated that the therapeutic outcome of carboplatin regimens is dependent on carboplatin plasma exposure. Low AUC values (< 4 mg/mL · min) were associated with a poor outcome for ovarian[10] and testicular[11] cancer. However, too high AUC values are most likely associated with unacceptable myelosuppression that may be life-threatening and would delay the following course and then possibly the efficacy.[12] To refine the target AUCs, we applied the model initially proposed by Friberg et al[6] and analyzed separately both platelet counts and ANCs from 338 to 383 patients treated with carboplatin. The present study is the largest to date in terms of the number of patients and the diversity of the evaluated covariates.

Before discussing the results of the covariates analyses, we emphasize that all the complementary evaluations performed to validate the final models indicated the agreement between the models and

Table 2. Final Covariate Models on the Whole Data Set

Final Model			
ANC (n = 338)		Platelet Counts (n = 383)	
Estimate	95% CI	Estimate	95% CI
Base = $\theta_1 \times (ALB/37)^{\theta 2} \times \theta_3^{CORT}$		Base = $\theta_1 \times \theta_2^{PTT}$	
θ_1 = 4.54	4.29 to 4.79	θ_1 = 345	332 to 358
θ_2 = −0.772	−1.041 to −0.503	θ_2 = 0.832	0.824 to 0.840
θ_3 = 1.34	1.155 to 1.525	IIV = 30.1%	27.0% to 32.9%
IIV = 36.2%	32.1% to 39.9%		
MTT = $\theta_4 \times \theta_5^{MONO}$		MTT = θ_3	
θ_4 = 150	142 to 158	θ_3 = 224	203 to 245
θ_5 = 1.87	1.541 to 2.199	IIV = 30.9%	25.9% to 35.1%
IIV = 20.7%	17.5% to 27.0%		
Slope = $\theta_6 \times \theta_7^{MONO}$		Slope = $\theta_4 \times \theta_5^{PACLI} \times \theta_6^{VP16} \times \theta_7^{GEM}$	
θ_6 = 0.808	0.742 to 0.874	θ_4 = 0.579	0.500 to 0.658
θ_7 = 0.569	0.455 to 0.680	θ_5 = 0.764	0.646 to 0.882
IIV = 24.1%	22.8% to 38.1%	θ_6 = 1.45	1.11 to 1.79
IOV = 20.2%	27.9% to 42.4%	θ_7 = 2.33	1.72 to 2.94
		IIV = 47.1%	38.7% to 54.2%
		IOV = 28.6%	21.7% to 34.2%
Power = θ_8		Power = θ_8	
θ_8 = 0.146	0.136 to 0.156	θ_8 = 0.233	0.227 to 0.239
(no IIV estimated)		IIV = 22.7%	15.8% to 28.0%
Residual variability = 61.6%	59.6% to 63.6%	Residual variability = 48.5%	47.0% to 49.9%

Alternative Models (independent deletion of each covariate)	
Deleted Covariate	ΔOBJ
Without ALB	+ 33
Without CORT	+ 27
Without MONO on MTT	+ 110
Without MONO on Slope	+ 39
Without PTT	+ 26
Without PACLI	+ 13
Without VP16	+ 10
Without GEM	+ 33

Abbreviations: ANC, absolute neutrophil count; ALB, albuminemia; CORT, multiple-dose corticosteroid administration; PTT, previous pretreatment (= 1) versus any (= 0); IIV, interindividual variability; MTT, mean transit time; MONO, carboplatin in monochemotherapy (= 1) versus in combination (= 0); PACLI, carboplatin in combination with paclitaxel (= 1) versus other combination or monochemotherapy (= 0); VP16, carboplatin in combination with etoposide; GEM, carboplatin in combination with gemcitabine; IOV, inter-occasion (inter-cycle) variability; ΔOBJ, difference in objective function between the final model and the model without the covariate (if ΔOBJ > +6.63, the covariate is significant with a *P* value of .01).

observations for both platelet counts and ANCs. An advanced internal validation was done by data splitting, and VPCs were performed to globally evaluate the models and ensure that the variability in the data was correctly described, together with the effects of the covariates. The results of this model evaluation were completely satisfactory. The large number of patients may explain the good precision of the estimation obtained for the model parameters, including the parameters related to covariate effects.

Importantly, none of the analyzed demographic, biologic, or pharmacogenetic covariates had a significant impact on hematopoietic toxicity sensitivity (slope). The results obtained in patients treated with standard carboplatin regimens showed that the main covariate for carboplatin myelosuppression is the nature of the associated chemotherapy (Fig 4). The results show that thrombocytopenia represents the dose-limiting toxicity for patients receiving carboplatin with gemcitabine or as monochemotherapy and that neutropenia is the dose-limiting toxicity in patients receiving the carboplatin-paclitaxel combination. Patients cotreated with other cytotoxic agents face a substantial risk of both neutropenia and thrombocytopenia (particularly for the carboplatin-etoposide regimen). Pretreatment was identified as a significant covariate of a lower basal platelet count and should be considered when the choice of the target carboplatin AUC is exclusively or partly dependent on the risk of thrombocytopenia (monochemotherapy and combination other than paclitaxel). A carboplatin AUC of 5 or 7 mg/mL · min was associated with a risk of grade 4 neutropenia of 21% or 42%, respectively, for any combination without multiple-dose corticosteroid administration, versus 14% or 32% for any combination with multiple-dose corticosteroid administration. In the case of monochemotherapy, an AUC of 5 or 7 mg/mL · min was associated with a 7% and 16% rate of grade 3 to 4 thrombocytopenia, respectively, for patients who were not pretreated; these figures are 9% or 22%, respectively, in pretreated patients. In combination with gemcitabine, an AUC of 5 was associated with 31% and 38% grade 3 to 4 thrombocytopenia for patients who were not pretreated and pretreated patients, respectively. In addition, simulation analysis showed that the carboplatin-gemcitabine regimen (with a

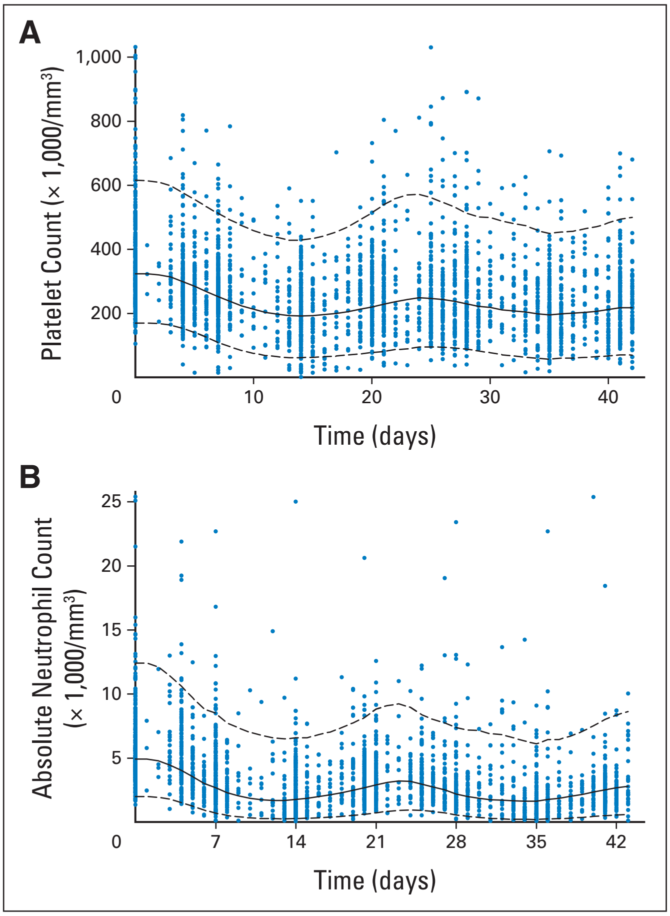

Fig 3. Visual predictive check for the final models obtained with the whole data set using 1,000 simulations of the study design. The solid line represents the median of the simulations, the dashed lines the fifth and 95th percent prediction intervals, and the dots represent the observed counts. (A) Platelet count; (B) absolute neutrophil count (two overlaid points at day 21 [35 × 1,000/μL] and day 22 [44 × 1,000/μL]).

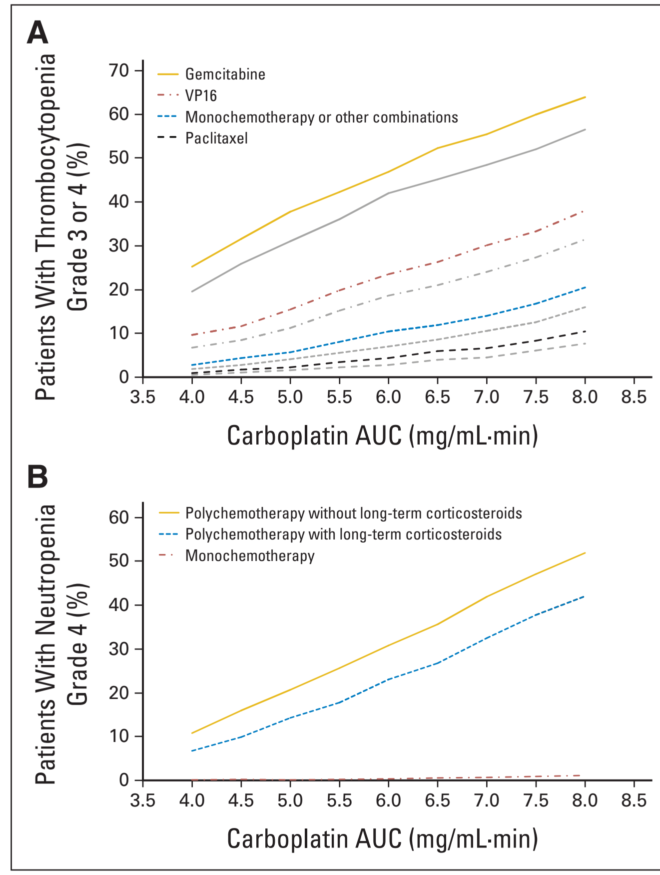

Fig 4. Simulated percentage of patients experiencing hematotoxicity according to the final models obtained with the whole data set. (A) Thrombocytopenia grade 3 or 4. Gold, red, and blue indicate previous chemotherapy; gray indicates no chemotherapy. (B) Neutropenia grade 4. VP16, etoposide; AUC, area under the curve.

carboplatin AUC of 5 mg/mL · min) was associated with a 44% lower basal level of platelet counts at day 21 (scheduled day for cycle 2) compared with day 0 (just before cycle 1). By comparison, the basal platelet level at cycle 2 was only 27% lower for monochemotherapy and combinations other than gemcitabine, etoposide, or paclitaxel compared with cycle 1. These last results illustrate the cumulative thrombocytopenia resulting from the use of carboplatin-gemcitabine regimen, which imposes a delay and/or a decrease in the target carboplatin AUC for the following cycles. By choosing an AUC of 5 mg/L · min, 35% of the patients cannot be treated at day 21 compared with only 15% at day 28, suggesting that a 4-week regimen should be recommended at this level of AUC.

Multiple-dose corticosteroid administration was associated with a higher basal level of ANC; this may reflect a consequence of the demargination of intravascular neutrophils. However, the inverse relationship observed between albuminemia and the basal ANC level is more difficult to explain.

Interestingly, the current data confirmed several clinical and preclinical observations showing a platelet-sparing effect of paclitaxel that resulted in a lower risk of thrombocytopenia after carboplatin administration.[13,14] The present PK-PD analysis enables this effect to be quantified: the slope value was 24% lower when carboplatin was combined with paclitaxel. Although not significant, there was also a trend for lower slope values in the ANC analysis when carboplatin was combined with paclitaxel in comparison with other combinations such as carboplatin and gemcitabine (Fig 2). These results suggest that a higher target AUC (eg, 6 mg/mL · ml rather than 5 mg/mL · min) may be chosen for the carboplatin-paclitaxel regimen. A limitation of this analysis was the absence of paclitaxel PK data. Another multi-institutional study published by Joerger et al[15] showed that the time above threshold paclitaxel concentration of 0.05 μmol/L was a good predictor for severe neutropenia. However, implementation of this result for individual dosing in clinical practice would require therapeutic drug monitoring, which could only be possible after completion of cycle 1. The advantage of PK-PD modeling related to carboplatin is that it can be applied in routine clinical practice because carboplatin clearance may be accurately predicted without any bias, particularly by considering cystatin C plasma levels together with usual patient characteristics (serum creatinine, body weight, age, and sex).[7]

Finally, several polymorphisms corresponding to genes coding for proteins involved in cellular detoxification of platinum or platinum adducts DNA repair have been investigated. The slope values did not differ significantly between pharmacogenetic subgroups. Although these results do not prove the absence of relationships, they are consistent with the pharmacogenetic assessment of the Scottish Randomized Trial in Ovarian Cancer[16] that did not reveal any significant association between genotype and observed toxicity. Age was also not identified as a significant pharmacodynamic covariate. However, it represents a significant pharmacokinetic characteristic that should be considered for the prediction of the individual carboplatin clearance and then in carboplatin individual dosing.

The additional benefit of the Friberg model is that both the duration of cytopenia, and the area under the curve of ANC or platelet count versus time corresponding to grade 4 or 3 to 4, respectively, can be determined. For example, for patients treated with a carboplatin AUC of 5 mg/mL · min experiencing grade 3 to 4 thrombocytopenia, the mean duration and area under the curve of a safe platelet count were 6.0 days and 120×10^9/L · day, respectively, when cotreated with gemcitabine. When cotreated with paclitaxel, the corresponding values were only 4.7 days and 53×10^9/L · day.

In conclusion, the methodology used to analyze these multicenter carboplatin data may be recommended in the development of new drugs, but also for other currently used drugs. Recently, Wallin et al[17] transferred a semi-mechanistic myelosuppression model from NONMEM to a dosing tool in Excel (Microsoft, Redmond, WA): The user provides the observed ANC for a patient from a previous treatment course and requests the dose that results in the desired nadir for the next course. We plan to further develop such a tool for individual carboplatin dosing in routine practice. Such a tool should take into account PK variability and both neutropenia and thrombocytopenia-related toxicity.

AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

The author(s) indicated no potential conflicts of interest.

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